

Three way handshake :-

The term given for process used to establish a connection b/w two devices. The 3way Handshake communicated using a few special messages. The table below highlights the main ones:

Step	Message	Description
1	SYN (Synchronize)	A SYN is initial packet sent by a client during handshake. This packet is used to initiate a connection and synchronize the two devices together.
2	SYN/Ack (Synchronize/Acknowledge)	This packet is sent by receiving device (Server) to acknowledge the Synchronisation attempt from the client.
3	Ack (Acknowledge)	The acknowledgement packet can be used by either the client or server to acknowledge that a series of messages/packets have been successfully received.
4	Data	Once a connection is established, data (such as byte of file) sent via the "Data" message.
5	Fin (Finish)	This packet is used to cleanly (properly) close the connection after it has been complete.
6	RST (Reset)	This packet abruptly ends all communication. This is the last resort and indicates there was problem during the process. For example, if service or application is not working properly, the system has faults such as low resources.

Simply

SYN (Synchronize) :- The client knocks on server door to say Hey, I want to start a connection.

SYN/ACK (Synchronize/Acknowledge) :- The server opens door and replies, 'Hey, I hear you, let's connect!'

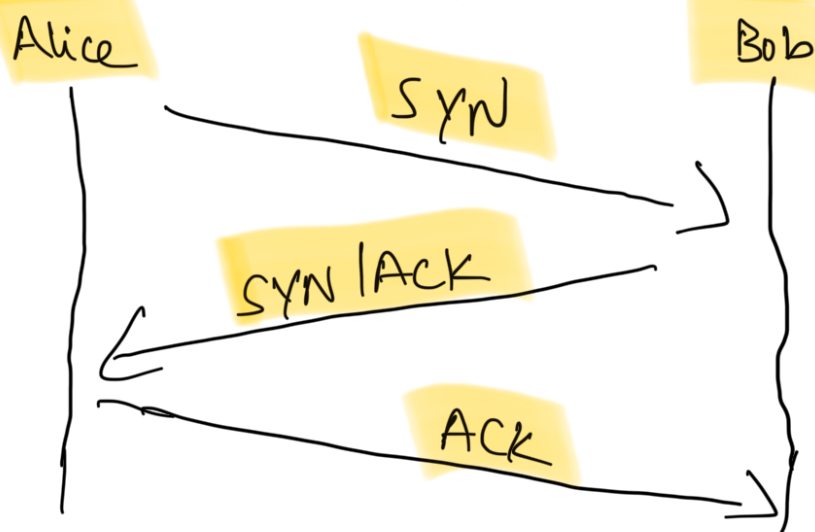
ACK (Acknowledge) :- The client says Awesome got it! That complete 3way handshake.

Data :- Now they are connected, they actually exchange information like downloading file or webpage.

Fin (Finish): - when they are done, they politely say goodbye and close connection cleanly.

RST (Reset): - If something goes wrong or crashes, this instantly cuts the connection off like hanging up phone abruptly.

* The diagram below shows a normal 3 way handshake process b/w Alice and Bob. In real life this would be between two devices.



* Any sent data is given a random number sequence and reconstructed using number sequence and incremented by 1.

Both computers must agree on same number sequence for data to be sent in correct order. This order agreed upon during 3 steps.

1. **SYN** - Client - here my initial sequence number (ISN) to synchronize with (0).

2. **SYN/ACK**: - Server - Here is my initial sequence number (ISN) to synchronize with (5,000) and I acknowledge your initial number sequence (0).

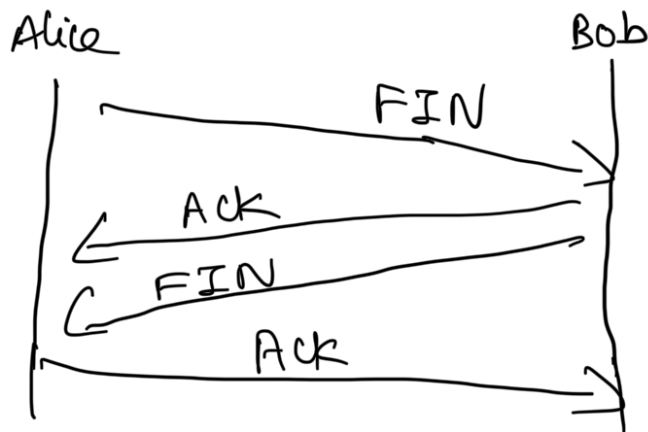
3. **ACK**: - Client - I Acknowledge your initial sequence number (ISN) of (5000), here is some data that is my ISN + 1 (0+1)

Device | Initial Number sequence (ISN) | Final Number sequence

Client (Sender)	0	$0+1=1$
Client (Sender)	1	$1+1=2$
Client (Sender)	2	$2+1=3$

TCP closing a connection: —

Alice and Bob



The Alice sent Bob as "FIN" packet. Because Bob received this, he will let Alice know that he received it and that he also wants to close connection (using Fin). Alice has heard Bob loud and clear and will let Bob know that she Acknowledged this.

Quiz:—

1) What is the header in TCP packet that ensures Integrity of data?

A) checksum

2) provide the order of normal three way handshake

A) SYN, SYN/ACK, ACK

Task 4 :- UDP/IP

*The user data gram protocol (UDP) is another protocol that is used to communicate data between devices.

* Unlike its brother TCP, UDP is stateless protocol that doesn't

require a constant connection between the two devices for data to be sent. For example, three way handshake does not occur, nor is there any synchronisation b/w two devices.

* UDP is used in situation where applications can tolerate data being lost (such as video streaming or voice chat) or in scenarios where unstable connection is not the end all.

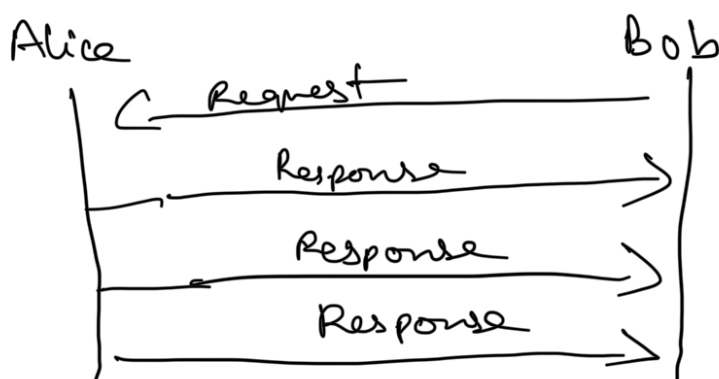
* UDP packets are much simpler than TCP packets have fewer headers.

Header	Description
Time to live (TTL)	This field setup expiry timer for packet, so it doesn't clog up your network if it never manages to reach a host or escape
Source Address	The IP address of device that packet is being sent from, so that data knows where to return to.
Destination Address	The devices IP address the packet is being sent to so that data knows where to travel next.
Source port	This value is port number that is opened by sender to send UDP packet from. This value is randomly chosen (0-65535) that aren't already in use at that time.
Destination port	This value is port number that an application or service is running on remote host. Ex: web browser running on port 80. Unlike Source port, the value is not chosen random.
Data	This header is where data, i.e. bytes of files that is being transmitted, is stored.

UDP is different from TCP. we should recall UDP is stateless.

No acknowledgment is sent during connection.

In real life Alice Bob, this would be b/w two devices



Quiz: —

1) what does UDP stand for?

A) user datagram protocol.

2) what type of connection is UDP?

A) stateless

3) what protocol would you use to transfer a file

A) TCP

4) what protocol would you use to have video call?

A) UDP